Equational proofs for a theorem by Jacobson for a significant density of even exponents

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Abstract

For each positive integer $n \geq 2$, a purely equational proof must exist for the statement that a ring R in which $x^n = x$ holds for all $x \in R$, must be commutative. By elementary means, we find such equational proofs for a significant density of even exponents n by establishing a reduction step that is as strong as can reasonably be expected by decreasing the given exponent n in a way that is minimal in view of the finite fields satisfying the same equation.

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1 Introduction

A classical result by Jacobson [Jac45] from 1945 states that if R is a ring with the property that for every $x \in R$, there exists some integer $n(x) \geq 2$ such that $x^{n(x)} = x$, then the ring R must be commutative. Here, R need not be unital. The standard proof can be found in §12 in Lam's *A First Course in Noncommutative Rings* [Lam91], in which the proof of Jacobson's result in its complete generality is reduced to little more than a single page of reasoning after applying Herstein's Lemma and subsequently appealing to various structure theorems for certain classes of rings.

However, various authors observed that to prove Jacobson's result, there is no need for such powerful tools, and more elementary proofs have since been found. Examples can be found in Herstein's paper [Her54], Rogers's work [Rog71] and [Rog72], and in Nagahara and Tominaga's very short paper [NT74]. These proofs are much simpler than Jacobson's original proof, but they do rely on various ring theoretic notions. In particular, they venture beyond the framework of pure equational logic.

In a first course on ring theory, it is not uncommon to challenge students to prove Jacobson's theorem in the special case that one can take $n(x) = 2$ for all $x \in R$. In other words, tasking them to show that all Boolean rings must be commutative, which follows from expanding the equation $(x + y)^2 = x + y$ and cancelling, combined with the observation that $x = (-x)^2 = -x$.

In various online spaces, numerous examples can be found of students tasked to prove Jacobson's theorem in the special case that one can take $n(x) = 3$ for all $x \in R$; a problem that is considerably less trivial but can still be done purely equationally with some patience. It is natural to consider the special case of Jacobson's theorem for an arbitrary fixed exponent $n(x) = n$; we call all such rings $J(n)$-rings. By Birkhoff's completeness theorem, included as Theorem 14.19 in [SB81], proofs using purely equational logic must exist in each case, but finding them and writing them down is a separate challenge. For $n \leq 6$, such proofs are easily found on the internet. Many have wondered about a more general solution to this problem, but often with little success.

A great source for explicit equational proofs for small exponents is Morita's article [Mor78], which is notoriously difficult to obtain but contains proofs for all $n \leq 25$ and all even $n \leq 50$, with the exception of $n \in \{16, 22, 32, 46\}$. In [Zha90], Zhang uses an idea similar to the one presented in the present note to obtain equational proofs for an *empirically* positive density, with the drawback that deciding whether or not their strategy works for a given n is computationally as expensive as finding the equational proof itself. Significant progress was made by Burris and Lawrence in [BL91] by describing explicitly a finite set of rewrite rules complete for the equational theory of finite fields. However, they rely on the general results from [Jac45] to relate the equational theory for $J(n)$-rings to that of all finite $J(n)$-fields to draw their conclusions.

In this short note we propose an elementary reduction step for general even exponents that yields equational proofs for Jacobson's commutativity theorem for a very significant positive density of even integers. This makes explicit the general reduction step from the equational theory for $J(n)$-rings to that of all finite $J(n)$-fields used in [BL91]. In this sense, our reduction step is as strong as one could expect.

2 The reduction step

To obtain our reduction, our strategy is as follows. Fix a positive even integer n and a $J(n)$-ring R. Note that we do not assume that R is unital. We may then notice that $x = (-x)^n = -x$ for all $x \in R$, and as such, the ring is of characteristic at most 2. For any $x \in R$ and any $h \in X\mathbb{F}_2[X]$, it must hold that $h(x)^n - h(x) = 0 \in R$. Let $I_n \subset \mathbb{F}_2[X]$ be the ideal spanned by all these relations. Then by construction, for any $f \in I_n$ and $x \in R$, we must have that $f(x) = 0 \in R$. Because $\mathbb{F}_2[X]$ is a principal ideal domain, it follows that $I_n = (g_n)$ for some $g_n \in \mathbb{F}_2[X]$. Our key result determines g_n exactly using only some very elementary algebra. We use the convention that $\mathbb{N} = \mathbb{Z}_{\geq 1}$.

Proposition 1. *With the definitions from above, define the set*

$$S_n = \{m \in \mathbb{N} : 2^m - 1 \mid n - 1\}.$$

Then we have

$$g_n = \operatorname{lcm}\left\{X^{2^m} - X \mid m \in S_n\right\}.$$

Proof. We use the principle that two squarefree polynomials over $\mathbb{F}_2$ are equal if and only if they have the same zeroes in an algebraic closure $\overline{\mathbb{F}}_2$ of $\mathbb{F}_2$. Indeed, both sides are squarefree because the polynomial $X^k - X$ is separable for any positive even integer k. The zeroes of the right hand side are easily determined; some $\alpha \in \overline{\mathbb{F}}_2$ is one of its zeroes if and only if it is a zero of $X^{2^m} - X$ for some $m \in S_n$, or equivalently, if and only if $\alpha \in \mathbb{F}_{2^m}$ for some $m \in S_n$. On the other hand, some $\alpha \in \overline{\mathbb{F}}_2$ is a zero of g_n if and only if it is a zero of all $h^n - h$ for $h \in X\mathbb{F}_2[X]$. We have thus reduced to showing that for $\alpha \in \overline{\mathbb{F}}_2$,

$$h(\alpha)^n = h(\alpha) \text{ for all } h \in X\mathbb{F}_2[X] \iff \alpha \in \mathbb{F}_{2^m} \text{ for some } m \in S_n.$$

If $\alpha \in \mathbb{F}_{2^m}$ for some $m \in S_n$ and $h(\alpha) = 0$, then we are done. If $h(\alpha) \in \mathbb{F}_{2^m}^\times$ instead, it holds that $h(\alpha)^{2^m-1} = 1$. By definition of m, raising this expression to some power yields that $h(\alpha)^{n-1} = 1$, proving one direction.

Now let $\alpha \in \overline{\mathbb{F}}_2$ and suppose that $h(\alpha)^n = h(\alpha)$ for all $h \in X\mathbb{F}_2[X]$. If we write $\mathbb{F}_2[\alpha] = \mathbb{F}_{2^m}$ for some $m \in \mathbb{N}$, we must show that $m \in S_n$. Using that $X^{2^m-1} \in X\mathbb{F}_2[X]$ is constantly equal to 1 on $\mathbb{F}_{2^m}^\times$, we observe that

$$\left\{h(\alpha) \mid h \in X\mathbb{F}_2[X]\right\} = \left\{h(\alpha) \mid h \in \mathbb{F}_2[X]\right\} = \mathbb{F}_2[\alpha] = \mathbb{F}_{2^m}.$$

In other words, we have shown that $\beta^n = \beta$ for all $\beta \in \mathbb{F}_{2^m}$. Since $\mathbb{F}_{2^m}^\times$ is cyclic of order $2^m - 1$, we may choose β to be a generator. It then follows immediately that $2^m - 1 \mid n - 1$, showing $m \in S_n$ and completing the proof. $\square$